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CausalMixGPD: Bayesian Nonparametric Modeling with Optional Heavy-Tail Augmentation and Causal Extensions

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Abstract

Heavy-tailed outcomes arise in settings where rare extreme events drive risk, cost, or policy decisions. Parametric models often describe the center of the distribution adequately but can miss extreme behavior, whereas fully nonparametric models lack principled tail extrapolation. We develop CausalMixGPD, an R package for Bayesian modeling of heavy-tailed outcomes that combines Dirichlet process mixtures for the bulk with optional generalized Pareto tails in a unified framework. The package supports unconditional and covariate-dependent mixture models, posterior computation through Markov chain Monte Carlo in nimble, and arm-specific outcome models for causal inference. The package reports posterior predictive summaries for densities, quantiles, means, survival curves, and causal contrasts, with particular emphasis on extreme quantiles and other tail-sensitive functionals. The same framework also supports predictor-dependent clustering under multiple kernel specifications. In practice, CausalMixGPD provides a practical and extensible workflow for heavy-tailed density estimation, clustering, and causal reporting.

Keywords: Bayesian nonparametrics; Dirichlet Process Mixture; Generalized Pareto Distribution; Extreme Value Theory; Cluster Analysis; Causal Inference; R; nimble.

1. Introduction

Causal inference is often summarized through average effects, but many applications require distributional comparisons between treatment regimes. In clinical trials, investigators may want to understand how a treatment changes outcomes across the outcome scale, including effects among individuals in the upper tail where events are rare but consequences can be substantial. This shift from mean-based to distribution-based causal questions makes causal analysis tightly linked to how well we can estimate (and predict) outcome distributions under

competing treatment assignments.

A growing R ecosystem supports Bayesian causal inference through flexible outcome modeling and posterior uncertainty quantification, including tree-based and model-averaging approaches such as bcf, bartCause, and BCEE (R Package Authors 2024a, 2025a, 2023a). These tools are effective for mean and heterogeneous effect estimation, but distributional reporting is more demanding: quantiles estimated separately can cross, and tail-focused contrasts require the full outcome distribution rather than a collection of marginal summaries. CausalMixGPD addresses this by modeling outcome densities directly and deriving quantiles, means, survival curves, and tail summaries from one posterior predictive object.

Dirichlet Process Mixtures (DPM) provide a versatile nonparametric solution for density estimation because they can capture multimodality and latent heterogeneity without fixing the number of mixing components in advance (Escobar and West 1995). In the R ecosystem, Bayesian nonparametric mixture modeling is well supported by software based on Dirichlet process (DP) ideas. dirichletprocess provides a flexible interface for specifying DPMs and fitting them by posterior simulation, yielding posterior inference for mixture weights, component labels, and predictive densities under user-chosen kernels (R Package Authors 2023b). The package DPpackage offers a broad suite of Bayesian semi- and nonparametric models in R, including DP-based formulations for marginal and conditional densities, regression functions, and model-based predictive distributions (Jara, Hanson, Quintana, Müller, and Rosner 2011). More recently, BNPmix emphasizes practical Bayesian nonparametric mixture approaches, providing posterior inference for clustering structure and density/regression estimation under DP and Pitman–Yor-type mixture specifications (Corradin, Canale, and Nipoti 2021).

That clustering perspective is also central to the present package. Because DP mixtures induce random partitions through latent allocation variables, they naturally support model-based subgroup discovery in addition to density estimation. Bayesian cluster analysis is typically reported through label-invariant summaries such as pairwise co-clustering probabilities (Binder 1978) and representative partitions selected by Dahl's least-squares criterion (Dahl 2006). In regression settings, clustering itself can depend on predictors: dependent Dirichlet process constructions allow covariates to affect mixture weights, component-specific parameters, or both (MacEachern 1999, 2000; De Iorio, Müller, Rosner, and MacEachern 2004; Quintana, Müller, Jara, and MacEachern 2022), and predictor-dependent stick-breaking schemes provide a practical route for covariate-modulated cluster prevalence (Ren, Du, Carin, and Dunson 2011). Existing software supports parts of this workflow, but typically focuses on bulk density estimation and clustering under a single kernel family without an explicit adjustment for heavy-tailed data. As a result, applications that require both latent subgroup discovery and tail extrapolation still need additional modeling outside the standard Bayesian nonparametric mixture workflow.

Estimating values far in the tail requires special care when data become scarce, and fitting a single model to heavy-tailed data can distort the fit in the central region. Extreme value theory (EVT) provides a principled framework for modeling and extrapolation of tail behavior. EVT characterizes the asymptotic distribution of threshold exceedances, which can be used to estimate tail probabilities and quantiles beyond the range of observed data (Balkema and de Haan 1974; Pickands 1975; Coles 2001). Bayesian implementations of EVT are available in R when posterior uncertainty quantification for tail quantities is required. evdbayes provides Bayesian inference for standard extreme-value models via posterior simulation, enabling posterior and posterior-predictive summaries for extreme-event quantities (R Package Authors

2025b). For peaks-over-threshold (POT) approaches, the package POT implements a focused collection of tools for GPD modeling and threshold-based tail analysis (R Package Authors 2024b). For extreme quantile estimation and rare-event tail summaries, extremefit provides a software implementation targeted at extreme-quantile estimation (Durrieu, Grama, Jaunatre, Pham, and Tricot 2018). Existing tools typically emphasize either flexible bulk modeling or dedicated tail analysis. Applications that need both predictor-dependent clustering and causal contrasts in the upper tail still require users to stitch together multiple workflows.

This article introduces an R package (CausalMixGPD) that connects Bayesian nonparametric modeling with EVT, clustering, and robust causal analyses in the presence of covariates (confounders). The package provides three simultaneous capabilities:

(i) **BNP density estimation with an optional GPD tail:** the package fits DPMs below a threshold and can splice a generalized Pareto tail above it for tail probabilities, extreme quantiles, and rare-event prediction.

(ii) **Predictor-dependent clustering:** dedicated clustering wrappers support dependence through weights, component parameters, or both, and report label-invariant summaries through posterior similarity matrices and Dahl representative partitions (Binder 1978; Dahl 2006).

(iii) **Causal estimation via posterior functionals:** treatment-specific outcome distributions are modeled directly, so average, restricted-mean, and quantile treatment effects are computed from the fitted distributions in randomized and observational studies, with optional propensity-score augmentation in the latter.

Overall, CausalMixGPD offers a unified workflow for density estimation, prediction, clustering, and causal reporting when tail behavior matters. The package is implemented in R, with computationally intensive components integrated through nimble (R Package Authors 2025c). Section 2 introduces the spliced model and the reported causal estimands. Section 3 summarizes the package workflow and user interface. Sections 4 and 5 illustrate the clustering and causal workflows, and Section 6 closes with limitations and future directions.

2. Background and Preliminaries

Let the outcome–covariate pair be defined by

$$(Y, X) \in \mathbb{R} \times \mathbb{R}^p. \tag{1}$$

For a fixed covariate value x, define

$$F(y \mid x) = \Pr\,(Y \le y \mid X = x), \qquad f(y \mid x) = \frac{\partial}{\partial y} F(y \mid x), \tag{2}$$

and

$$Q(\tau \mid x) = \inf\Big\{y \in \mathbb{R} :\, F(y \mid x) \ge \tau\Big\}, \qquad 0 < \tau < 1. \tag{3}$$

CausalMixGPD treats $F(\cdot \mid x)$ as the primary inferential object. The center of the distribution is modeled nonparametrically, and the upper tail can be replaced by a generalized Pareto

component when extreme-value extrapolation is needed. Reported quantities such as densities, survival curves, quantiles, means, and causal contrasts are posterior functionals of this fitted distribution.

2.1. Generalized Pareto Distribution for Threshold Exceedances

In modeling terms, the tail region is defined by values that exceed a high threshold. Extreme value theory gives a standard approximation for exceedances over such a threshold. Let $u(x)$ be a high threshold, which may depend on x, and define the exceedance

$$Z = Y - u(x), \tag{4}$$

considered conditionally on $Y > u(x)$. The conditional excess distribution is given by

$$F_u(z \mid x) = \Pr\big(Y - u(x) \le z \mid Y > u(x), X = x\big), \qquad z \ge 0. \tag{5}$$

Under broad regularity conditions, $F_u(\cdot \mid x)$ is well-approximated by a generalized Pareto distribution (GPD) when $u(x)$ is large (Balkema and de Haan 1974; Pickands 1975; Coles 2001).

For $y > u(x)$, the corresponding GPD cumulative distribution function is

$$F_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} 1 - \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0, \end{cases} \tag{6}$$

with support $y \ge u(x)$ when $\xi \ge 0$, and $u(x) \le y \le u(x) - \sigma(x)/\xi$ when $\xi < 0$. The corresponding density function is

$$f_{\mathrm{GPD}}(y \mid u(x), \sigma(x), \xi) = \begin{cases} \dfrac{1}{\sigma(x)} \left(1 + \xi \dfrac{y - u(x)}{\sigma(x)}\right)^{-1/\xi - 1}, & \xi \neq 0, \\ \dfrac{1}{\sigma(x)} \exp\left(-\dfrac{y - u(x)}{\sigma(x)}\right), & \xi = 0. \end{cases} \tag{7}$$

The associated GPD quantile function takes the form

$$Q_{\mathrm{GPD}}(\tau \mid u(x), \sigma(x), \xi) = \begin{cases} u(x) - \sigma(x) \log(1 - \tau), & \xi = 0, \\ u(x) + \dfrac{\sigma(x)}{\xi}\Big((1 - \tau)^{-\xi} - 1\Big), & \xi \neq 0, \end{cases} \qquad 0 < \tau < 1. \tag{8}$$

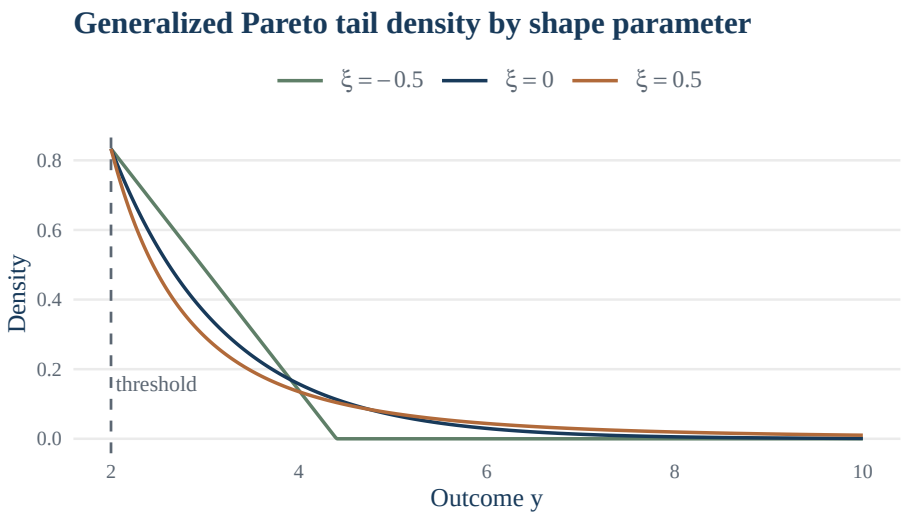

Figure 1: GPD tail density above threshold for multiple shape parameters.

Figure 1 illustrates the GPD density for different shape parameters ξ with a fixed scale parameter σ and threshold u. The shape parameter ξ controls the tail heaviness: $\xi < 0$ corresponds to a light tail with a finite upper bound, $\xi = 0$ corresponds to an exponential tail, and $\xi > 0$ corresponds to a heavy tail with infinite support. The GPD provides a flexible family for modeling exceedances above a high threshold, making it the tail component used in the spliced model.

2.2. Dirichlet Process Mixture Model for Conditional Densities

In this package, the conditional density of Y given X below a specific threshold $u(x)$ is modeled using a DPM regression model. Let $k(\cdot \mid x; \theta)$ be a user-chosen kernel family indexed by parameter θ. The hierarchical specification is

$$Y \mid X = x, \theta \sim k(\cdot \mid x; \theta), \qquad \theta \mid H \sim H, \qquad H \mid \kappa, H_0 \sim \mathrm{DP}(\kappa, H_0), \tag{9}$$

where H is an unknown mixing distribution on the kernel-parameter space, H_0 is a base measure, and $\kappa > 0$ is a concentration parameter (Ferguson 1973). Marginalizing H yields an (almost surely) discrete mixing distribution together with an infinite mixture representation of the form

$$f_{\mathrm{DP}}(y \mid x) = \int k(y \mid x; \theta)\, dH(\theta) = \sum_{j=1}^{\infty} w_j\, k(y \mid x; \theta_j), \qquad w_j \geq 0, \qquad \sum_{j=1}^{\infty} w_j = 1. \tag{10}$$

One representation for the weights is Sethuraman's stick-breaking construction (Sethuraman 1994), in which

$$w_1 = V_1, \qquad w_j = V_j \prod_{\ell<j} (1 - V_\ell), \qquad V_j \overset{\text{iid}}{\sim} \mathrm{Beta}(1, \kappa), \qquad \theta_j \overset{\text{iid}}{\sim} H_0. \tag{11}$$

An equivalent characterization is obtained by marginalizing over H, yielding the Blackwell–MacQueen Pólya urn representation. In particular, the conditional distribution of a parameter

value given the others can be written as

$$\theta_i \mid \{\theta_j\}_{j\neq i} \sim \frac{\kappa}{\kappa+n} H_0(\cdot) + \frac{1}{\kappa+n} \sum_{j=1}^{n} \delta_{\theta_j}(\cdot), \tag{12}$$

where $\delta_{\theta_j}(\cdot)$ is a point mass assigned to θ_j. This induces a clustering structure often described via the Chinese restaurant process (CRP) (Neal 2000). Both stick-breaking and CRP representations are widely used for posterior simulation in DPM models (Escobar and West 1995; Neal 2000). DPMs work without specifying the number of mixture components in advance and can capture complex distributional features such as multimodality, skewness, and heteroscedasticity by choosing an appropriate kernel family, making them a natural choice for modeling the bulk of the conditional distribution where data are plentiful and complex features may be present.

2.3. Spliced DPM–GPD Conditional Distribution

CausalMixGPD uses a spliced (bulk–tail) construction that retains the DPM fit below a high threshold and replaces the upper tail by a GPD component (Frigessi, Haug, and Rue 2002; Behrens, Lopes, and Gamerman 2004; Coles 2001). For a fixed covariate value x, let $u(x)$ denote the threshold, let $\sigma(x) > 0$ denote the tail scale above that threshold, and let $\xi \in \mathbb{R}$ denote the tail-shape parameter. Let Θ collect the bulk-mixture parameters and let Φ collect the tail quantities $\{u(x), \sigma(x), \xi\}$. Write $F_{\mathrm{DP}}(y \mid x; \Theta)$ for the bulk conditional CDF and $F_{\mathrm{GPD}}(y \mid x; \Phi)$ for the exceedance CDF. Then

$$p_u(x) = F_{\mathrm{DP}}(u(x) \mid x; \Theta) \tag{13}$$

is the fitted bulk mass below the threshold $u(x)$, and the spliced CDF is

$$F(y \mid x; \Theta, \Phi) = \begin{cases} F_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ p_u(x; \Theta) + \{1 - p_u(x; \Theta)\}\, F_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{14}$$

This construction is continuous at $u(x)$. Differentiating (14) gives

$$f(y \mid x; \Theta, \Phi) = \begin{cases} f_{\mathrm{DP}}(y \mid x; \Theta), & y \leq u(x), \\ \{1 - p_u(x; \Theta)\}\, f_{\mathrm{GPD}}(y \mid x; \Phi), & y > u(x), \end{cases} \tag{15}$$

where the tail density is rescaled by $1 - p_u(x; \Theta)$ so that $f(\cdot \mid x; \Theta, \Phi)$ integrates to one. Figure 2 gives a compact visual comparison between a one-component gamma density, a two-component gamma-mixture density, and a two-component gamma-mixture-plus-GPD spliced density with the splice set at the 80% bulk quantile.

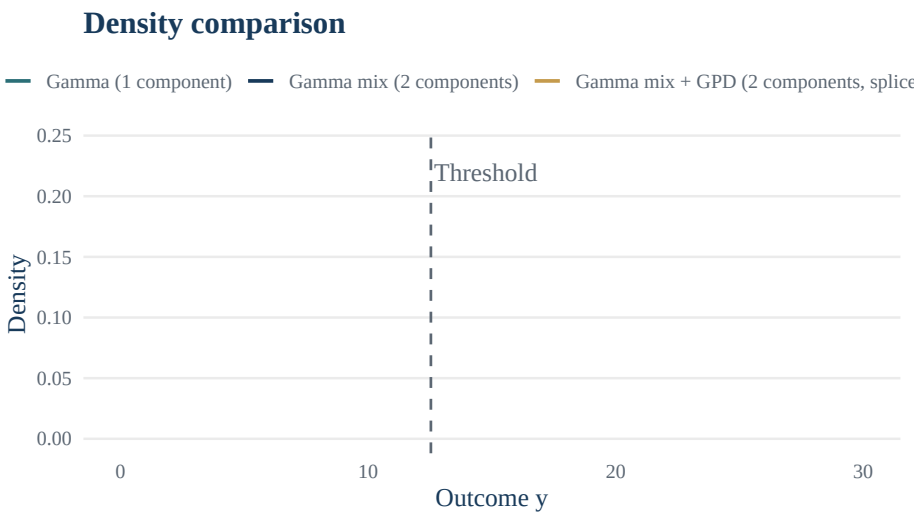

Figure 2: Comparison of three package-based curves: Gamma (1 component), Gamma mixture (2 components), and Gamma mixture + GPD (2 components) with splice threshold set at the 80th percentile of the bulk mixture.

For $\tau \leq p_u(x;\Theta)$, quantiles are obtained by inverting the bulk CDF. Let $Q_{\mathrm{DP}}(\tau \mid x;\Theta)$ denote that bulk quantile function. For $\tau > p_u(x;\Theta)$, the probability index must be rescaled relative to the exceedance probability, giving the rescaled index

$$\tilde{\tau}(x;\Theta) = \frac{\tau - p_u(x;\Theta)}{1 - p_u(x;\Theta)} \in (0,1), \tag{16}$$

The GPD quantile is

$$Q_{\mathrm{GPD}}(\tau \mid x;\Phi) = \begin{cases} u(x) - \sigma(x)\log(1-\tau), & \xi = 0, \\ u(x) + \dfrac{\sigma(x)}{\xi}\{(1-\tau)^{-\xi} - 1\}, & \xi \neq 0, \end{cases} \qquad 0 < \tau < 1.$$

Combining both cases, the spliced quantile function is

$$Q(\tau \mid x;\Theta,\Phi) = \begin{cases} Q_{\mathrm{DP}}(\tau \mid x;\Theta), & \tau \leq p_u(x;\Theta), \\ Q_{\mathrm{GPD}}(\tilde{\tau}(x;\Theta) \mid x;\Phi), & \tau > p_u(x;\Theta). \end{cases} \tag{17}$$

In later sections, Θ indexes the bulk DPM and Φ indexes the tail component. The GPD and DPM specifications are introduced in Sections 2.1 and 2.2 above.

2.4. Clustering under predictor-dependent Bayesian nonparametric mixtures

A common Bayesian nonparametric approach to conditional density modeling represents the collection $\{f(\cdot \mid x) : x \in \mathcal{X}\}$ through predictor-indexed random probability measures. Within the Dependent Dirichlet Process (DDP) framework of MacEachern (1999, 2000), one specifies a family $\{G_x : x \in \mathcal{X}\}$ of random mixing measures whose dependence across x induces flexible borrowing of information in $f(\cdot \mid x)$.

A generic mixture representation is

$$f(y \mid x) \;=\; \int k(y \mid x, \theta)\; dG_x(\theta), \tag{18}$$

where $k(\cdot \mid x, \theta)$ is a kernel and G_x is a predictor-indexed mixing measure. When G_x is discrete (as under stick-breaking constructions), latent parameters exhibit ties, inducing a random partition of observations. Introduce latent cluster-label variables z_i and component parameters $\{\theta_j\}_{j\geq 1}$ such that

$$y_i \mid z_i, \{\theta_j\}, x_i \sim k(\,\cdot \mid x_i, \theta_{z_i})\,. \tag{19}$$

Clustering is then characterized by the posterior distribution of the induced partition $\Pi = \{z_1, \ldots, z_n\}$ and summarized in a label-invariant manner using co-clustering functionals (Binder 1978; Dahl 2006).

In the clustering component considered here, we allow three forms of predictor dependence, each corresponding to a particular construction of G_x. In all cases, the conditional density admits the series form

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,, \tag{20}$$

with $w_j(x) \geq 0$ and $\sum_{j=1}^{\infty} w_j(x) = 1$ for all x.

- **Weight dependence only:** predictors modulate mixture weights while atoms are shared across x,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid \theta_j)\,. \tag{21}$$

  This is appropriate when covariates change cluster prevalence but not the within-cluster kernel shape. A common approach uses predictor-dependent stick-breaking, where stick lengths are transformed regressions in x; the logistic stick-breaking process constructs these lengths via logistic regressions and maps them to weights (Ren *et al.* 2011). Such models can be viewed as practical DDP instances in which dependence is introduced through smoothly varying weights (MacEachern 2000; Quintana *et al.* 2022).

- **Atom dependence only:** predictors enter through component-specific kernel parameters while weights remain global,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j\, k(y \mid x, \theta_j)\,. \tag{22}$$

  Here clusters correspond to distinct regression surfaces while w_j retains the interpretation of global prevalence. This structure aligns with DDP constructions that share weights while allowing predictor-indexed atoms (MacEachern 1999, 2000), and is closely related to ANOVA-type dependent priors for structured collections of random measures (De Iorio *et al.* 2004).

- **Weight and atom dependence:** the most flexible specification allows predictors to influence both prevalence and within-cluster behavior,

$$f(y \mid x) \;=\; \sum_{j=1}^{\infty} w_j(x)\, k(y \mid x, \theta_j)\,. \tag{23}$$

This corresponds to DDP-type models in which both selection probabilities and component-specific conditional structure vary with predictors (Quintana *et al.* 2022).

Posterior clustering summaries are reported using label-invariant functionals of Π. The posterior similarity matrix (PSM) has entries

$$S_{i\ell} \;=\; \Pr(z_i = z_\ell \mid \text{data}), \qquad i, \ell = 1, \ldots, n, \tag{24}$$

estimated by Monte Carlo averages over MCMC output. A representative partition is obtained by selecting the sampled partition whose adjacency matrix is closest to the PSM under squared loss (Dahl 2006), yielding a stable point estimate while retaining the PSM as the primary uncertainty summary. Appendix Section 3.3 provides further details on the package's implementation of these summaries.

2.5. Causal Inference: Notation, Identification, and Reported Estimands

We use the potential outcomes framework (Neyman 1990; Rubin 1974; Imbens and Rubin 2015). Let $A \in \{0, 1\}$ denote a binary treatment indicator ($A = 1$ treated, $A = 0$ control), let X denote pre-treatment covariates, and let $Y(a)$ denote the potential outcome under arm $a \in \{0, 1\}$. The observed outcome satisfies

$$Y = A\,Y(1) + (1 - A)\,Y(0). \tag{25}$$

The treatment-specific conditional distribution functions are the fundamental inferential objects of the framework. For each arm $a \in \{0, 1\}$, the arm-specific conditional CDF is

$$F_a(y \mid x) = \Pr\,\{Y(a) \le y \mid X = x\}, \qquad a \in \{0, 1\}. \tag{26}$$

In this package, the corresponding conditional densities $f_a(y \mid x)$ assume the spliced DPM structure in (15). The conditional τ-quantile is obtained by inverting $F_a(\cdot \mid x)$,

$$Q_a(\tau \mid x) = \inf\{y :\, F_a(y \mid x) \ge \tau\}, \qquad \tau \in (0, 1), \tag{27}$$

and the conditional mean is a functional of the conditional density,

$$\mu_a(x) = \mathrm{E}\{Y(a) \mid X = x\} = \int y\, f_a(y \mid x)\, dy. \tag{28}$$

Marginal (standardized) distributions are obtained by averaging conditional distributions over a covariate distribution. Writing F_a^{m} for the population-standardized arm-a CDF and F_a^{t} for the treated-standardized analog, these are defined by

$$F_a^{\mathrm{m}}(y) = \mathrm{E}\{F_a(y \mid X)\}, \qquad F_a^{\mathrm{t}}(y) = \mathrm{E}\{F_a(y \mid X) \mid A = 1\}. \tag{29}$$

The associated marginal quantile functions are obtained by inverting these standardized CDFs,

$$Q_a^{\mathrm{m}}(\tau) = \inf\{y :\, F_a^{\mathrm{m}}(y) \ge \tau\}, \qquad Q_a^{\mathrm{t}}(\tau) = \inf\{y :\, F_a^{\mathrm{t}}(y) \ge \tau\}, \tag{30}$$

while the marginal means integrate the conditional mean over the respective covariate distributions,

$$\mu_a^{\mathrm{m}} = \mathrm{E}\{\mu_a(X)\}, \qquad \mu_a^{\mathrm{t}} = \mathrm{E}\{\mu_a(X) \mid A = 1\}. \tag{31}$$

Identification of these quantities depends on the study design. In randomized experiments, randomization together with positivity implies exchangeability between treatment assignment and potential outcomes (Imbens and Rubin 2015). In observational studies, we rely on the following conditions (Rosenbaum and Rubin 1983; Imbens and Rubin 2015).

- **Stable Unit Treatment Value Assumption (SUTVA):** there is no interference between units and no hidden versions of treatment.
- **Consistency:** if $A = a$, then the observed outcome equals the corresponding potential outcome, that is, $Y = Y(a)$.
- **Conditional ignorability:** treatment assignment is as-if randomized given covariates, so that
$$\big(Y(0), Y(1)\big) \;\perp\; A \;\mid\; X. \tag{32}$$
- **Overlap (positivity):** every covariate profile has a strictly positive probability of receiving either treatment, meaning
$$0 < \Pr\,(A = 1 \mid X = x) < 1 \tag{33}$$
for all covariate values of interest.

Under these conditions, the treatment-specific conditional distributions are identified from the observed data as

$$F_a(y \mid x) = \Pr\,(Y \le y \mid A = a, X = x), \qquad a \in \{0, 1\}. \tag{34}$$

In observational settings, the package uses the propensity score (PS) to support the design stage (Rosenbaum and Rubin 1983). The PS is the conditional probability of treatment assignment given covariates,

$$\rho(x) = \Pr\,(A = 1 \mid X = x), \tag{35}$$

and a key property of the true propensity score is that conditioning on it balances covariates across treatment groups. Following a two-stage strategy, the package first estimates $\rho(x)$ using only (A, X) and then augments the outcome model covariates with the estimated score, so that the augmented predictor vector is

$$r(x) = \big(x^{\mathsf{T}}, \widehat{\rho}(x)\big)^{\mathsf{T}}. \tag{36}$$

The conditional outcome distributions are then modeled separately within each treatment arm as functions of the augmented predictor $r(x)$, and causal contrasts are computed as functionals of the estimated treatment-specific distributions. Figure 3 illustrates two treatment-specific conditional densities at a representative covariate profile. The dotted verticals mark the 90th conditional quantile for each arm, and the double-headed arrow gives the conditional quantile treatment effect (CQTE) at that level.

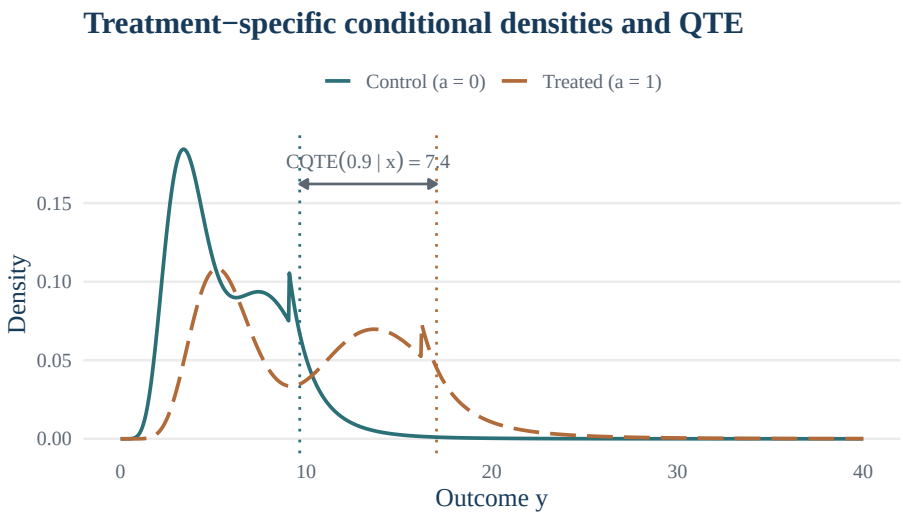

Figure 3: Treatment-specific conditional densities at a representative covariate profile. Dotted verticals mark the 90th quantile for each arm; the double-headed arrow gives the conditional quantile treatment effect (CQTE) at that level.

The package reports the causal estimands in Table 1. Here superscripts m and t denote full-population and treated-population standardization, respectively.

Table 1: Causal estimands reported by the package

Estimand	Formula	Target population
ATE	$\mu_1^{\mathrm{m}} - \mu_0^{\mathrm{m}}$	Entire population
ATT	$\mu_1^{\mathrm{t}} - \mu_0^{\mathrm{t}}$	Treated sub-population
QTE(τ)	$Q_1^{\mathrm{m}}(\tau) - Q_0^{\mathrm{m}}(\tau)$	Entire population at index τ
QTT(τ)	$Q_1^{\mathrm{t}}(\tau) - Q_0^{\mathrm{t}}(\tau)$	Treated sub-population at index τ
CATE(x)	$\mu_1(x) - \mu_0(x)$	Individual with covariate profile x
CQTE($\tau \mid x$)	$Q_1(\tau \mid x) - Q_0(\tau \mid x)$	Individual with x at index τ

The next section describes how the package fits the treatment-specific conditional distributions and computes these estimands from the posterior predictive output.

3. Package Overview: Workflow, Model, and Inference

This section briefly describes the basic workflow in using CausalMixGPDfor conditional modeling and prediction in a one-arm setup with a generalized Pareto distribution in the tail above a covariate-dependent threshold, while the causal version of this extends the exact same modeling, estimation, and prediction process with a treatment-specific outcome model and a propensity score component. Simulated data sets provided in this package are used to demonstrate the usage of different components in this package, and these data sets are provided in `data()`.

3.1. Spliced DPM–GPD model, likelihood, and priors

The observed data consist of n outcome–covariate pairs $\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n$, where $x_i \in \mathbb{R}^p$ denotes the covariate vector for unit i, optionally including an intercept column.

```
data("nc_posX100_p3_k2", package = "CausalMixGPD")
overview_dat <- data.frame(y = nc_posX100_p3_k2$y, nc_posX100_p3_k2$X)
```

The target conditional distribution $F(\cdot \mid x)$ and its spliced DPM–GPD construction are defined in Section 2, with the spliced CDF and density given in (14)–(15). For posterior computation, CausalMixGPD approximates the infinite bulk DPM by a finite truncation to J mixture components, so that

$$f_{\mathrm{DP}}(y \mid x; \Theta) \;\approx\; \sum_{j=1}^{J} w_j \, k(y \mid x; \theta_j),$$

where $j = 1, \ldots, J$ indexes components, each weight satisfies $w_j \geq 0$, and the weights sum to one up to a small residual mass assigned to the remaining components. Kernel parameters may be constant, component-specific, or covariate-dependent through link specifications of the form

$$g(\psi_j(x)) = x^\top \beta_{j,\psi}, \tag{37}$$

where $\psi_j(x)$ denotes a generic kernel parameter for component j at covariate value x and $g(\cdot)$ is a link function that enforces the parameter support, such as a log link for positive scale parameters. All code examples in this section use the MCMC settings shown below, which are intentionally short for illustration; longer chains and multiple chains are recommended for substantive analyses.

```
data.frame(mcmc_fixed)


  niter nburnin thin nchains seed
1   250      50    1       1 2026
```

The one-arm spliced model can be fit by inputting the formula, data, and control arguments to `dpmgpd()`. The `formula` argument follows standard R syntax and dictates which covariates to include in both the bulk and tail components. The `backend` argument specifies which representation of the DPM weights to use in the model. The option `"sb"` follows the stick breaking representation in equation (eq:sb), while the option `"crp"` follows the Chinese restaurant process representation. The `kernel` argument dictates the family of kernels used in the bulk, `components` dictates the truncation, and `mcmc` dictates the tuning.

```
overview_fit <- dpmgpd(formula = y ~ x1 + x2 + x3, data = overview_dat,
                       mcmc = mcmc_fixed)
```

Using the spliced density mentioned in (15), the likelihood is

$$L(\Theta, \Phi; \mathcal{D}) = \prod_{i=1}^{n} f(y_i \mid x_i; \Theta, \Phi), \tag{38}$$

where Θ collects the bulk parameters $\{w_j, \theta_j\}_{j=1}^{J}$, and Φ collects the tail parameters governing $u(x)$, $\sigma(x)$, and ξ. Equivalently, the log-likelihood can be written as

$$\begin{aligned}\log L(\Theta, \Phi; \mathcal{D}) \;=\; & \sum_{i=1}^{n}(1-\delta_i)\,\log f_{\mathrm{DP}}(y_i \mid x_i; \Theta) \\ & + \sum_{i=1}^{n}\delta_i\Big[\log\{1 - F_{\mathrm{DP}}(u(x_i) \mid x_i; \Theta)\} + \log f_{\mathrm{GPD}}(y_i \mid x_i; \Phi)\Big], \end{aligned} \tag{39}$$

where f_{GPD} is the GPD tail density from Section 2.3 and $F_{\mathrm{DP}}(\cdot \mid x)$ is the bulk mixture CDF. If the GPD tail is not included, then we have $f(y \mid x; \Theta, \Phi) \equiv f_{\mathrm{DP}}(y \mid x; \Theta)$, and we have a mixture over the bulk alone; a convenience interface to this bulk-only model, similar to `dpmgpd()`, is provided by the `dpmix()` wrapper.

```
overview_fit <- dpmix(formula = y ~ x1 + x2 + x3,  data = overview_dat,
                      mcmc = mcmc_fixed)
```

The prior distributions for the parameters Θ and Φ can be set by the user, with the defaults being weakly informative. For the main parameters of the DPM model, Θ, the default prior setting is to use stick-breaking priors for the mixture weights and conjugate priors for the kernel parameters when conjugacy holds. For the parameters Φ, the default setting is to use the regression-based form for the threshold $u(x)$ and scale $\sigma(x)$, together with a weakly informative prior for the shape parameter ξ.

3.2. Inference and prediction

Posterior simulation targets the joint posterior distribution of the bulk and tail parameters given data, which is

$$p(\Theta, \Phi \mid \mathcal{D}) \propto L(\Theta, \Phi; \mathcal{D})\, p(\Theta, \Phi)$$

using the likelihood (38) and the prior structure in Section 3.1. CausalMixGPD runs MCMC through nimble (R Package Authors 2025c). The fitted object supports `summary()` for posterior summaries, `plot()` for standard MCMC diagnostics, and `params()` for a lightweight parameter extractor.

```
summary(overview_fit)
params(overview_fit)
plot(overview_fit, family = c("traceplot"),
     params = "alpha")  # DPM concentration parameter
```

The package provides posterior summary of various functionals of the fitted conditional distribution in Section 2, as the estimation framework. Here in this manuscript, we focus on density $f(y \mid x)$, survival $S(y \mid x) = 1 - F(y \mid x)$, quantiles $Q(\tau \mid x)$, and mean $E(y \mid x)$. For types `density`, `survival`, and `quantiles`, the package evaluates each functional draw by draw from (14)–(17) and then summarizes the retained draws. For `type="mean"` and `type="rmean"`, posterior predictive Monte Carlo is employed in each draw; `type="rmean"` which calculates a restricted mean for a given cutoff is computationally stable even when the ordinary mean is unstable due to posterior mass near $\xi \geq 1$. The following code snippet is an example of how one can compute posterior density, quantiles and means at observed covariate values, along with credible intervals for quantiles and HPD intervals for means.

```
predict(overview_fit, type = "density",
  interval = "credible", level = 0.95)
predict(overview_fit, type = "quantile",
  index = c(0.25, 0.5, 0.75))
predict(overview_fit, type = "mean",
  interval = "hpd", level = 0.9)
```

Predictions at observed covariate values can be obtained by calling `predict()` with a dataset or profiles with a similar configuration to that used for model training.

For a new covariate value $x^\star$, prediction is based on the posterior predictive distribution

$$p(y^\star \mid x^\star, \mathcal{D}) = \int f(y^\star \mid x^\star; \Theta, \Phi)\, p(\Theta, \Phi \mid \mathcal{D})\, d\Theta\, d\Phi.$$

Density and survival predictions require a grid of y values supplied via the `y` argument, whereas quantile predictions require only the desired probability levels via `index`. The code snippet below demonstrates posterior survival and quantile predictions at new covariate profiles constructed from the empirical quartiles of the training predictors.

```
predict(overview_fit, newdata = overview_x_new,
  y = overview_y_grid, type = "survival", level = 0.95)
predict(overview_fit, newdata = overview_x_new, type = "quantile",
  index = c(0.5, 0.99), interval = "credible")
```

3.3. Clustering extension

```
data("nc_realX100_p3_k2", package = "CausalMixGPD")
dat_cl <- data.frame(y = nc_realX100_p3_k2$y[1:100],
  nc_realX100_p3_k2$X[1:100,])
```

The clustering framework reinterpret the fitted mixture as a latent partition and return cluster summaries rather than only density functionals. THe function `dpmix.cluster()` fits the bulk-only model and `dpmgpd.cluster()` adds the optional GPD tail for each mixture component, with predictor dependence entering through the weights, component parameters, or both. Since mixture labels are permutation-invariant, the main outputs are label-invariant summaries: the posterior similarity matrix (PSM), a representative partition, cluster sizes, and assignment scores (Binder 1978; Dahl 2006). The same `predict()` interface handles both training labels and new-observation assignment.

A bulk-only clustering model with covariate dependent parameters is fitted in the code snippet below, where `type="param"` specifies that predictors enter through the component-specific kernel parameters.

```
fit_cluster <- dpmix.cluster(
  y ~ x1 + x2 + x3,
  data       = dat_cl[1:100,],
  kernel     = "normal",
  type       = "param",
  components = 10,
  mcmc = mcmc_fixed
)
```

The same `predict()` interface returns either the posterior similarity matrix (PSM) or cluster labels depending on the `type` argument.

```
predict(fit_cluster, type = "psm")
```

New-observation cluster labels, together with per-cluster assignment scores, are obtained in the snippet below.

```
predict(fit_cluster, newdata = dat_cl[101:110,], type = "label")
```

As discussed in Section 2.4, the package supports three dependence patterns and reports clustering through PSM-based summaries and representative partitions.

3.4. Causal extension and propensity score augmentation

The causal framework fit separate outcome models for each treatment arm and can optionally augment the outcome covariates with a propensity-score summary. When propensity-score augmentation is requiredd (Observational study), treatment assignment is modeled as

$$A_i \mid \boldsymbol{x}_i \sim \text{Bernoulli}\big(e_{\boldsymbol{\gamma}}(\boldsymbol{x}_i)\big),$$

and the estimated propensity score is appended to the covariate vector to form the augmented predictor

$$\boldsymbol{r}(\boldsymbol{x}) = \Big(\boldsymbol{x}^\top,\ \psi(\widehat{e}(\boldsymbol{x}))\Big)^\top,$$

where the PS link (logit or probit) is specified through the `PS` argument and the transformation $\psi(\cdot)$ applied to the estimated score is controlled by `ps_scale` with default being identity. The outcome model for each arm is then a spliced DPM–GPD regression on the augmented covariates, reusing exactly the same mixture-regression and GPD-tail machinery as the one-arm workflow described in Section 3.1. For randomized experiments, the PS component is not needed and the same modeling workflow is applied to each arm without PS augmentation.

The code below loads a simulated benchmark dataset and assembles the data frame expected by the causal wrappers, with columns for the outcome, treatment indicator, and covariates.

The call to `dpmgpd.causal()` fits arm-specific spliced mixture models using the same structure as `dpmgpd()`, with additional arguments controlling the propensity-score model (`PS`, `ps_scale`, `ps_summary`) and separate MCMC settings for the outcome and PS stages (`mcmc_outcome`, `mcmc_ps`).

```
overview_causal_fit <- dpmgpd.causal(
  formula = y ~ x1 + x2 + x3,
  data = overview_causal_df,
  treat = "A",
  backend = "crp",
  kernel = "laplace",
  components = 5,
  PS = "logit",
  ps_scale = "logit",
  ps_summary = "mean",
  mcmc_outcome = mcmc_fixed,
  mcmc_ps = mcmc_fixed
)
```

Population-standardized estimates can be obtained using the functions `ate()` and `qte()`, while treated standardized versions are available via the functions `att()` and `qtt()`. Uncertainty quantification is available through intervals, and can be visualized by plotting the returned objects with `type="effect"` to show the estimated treatment effect curve across quantiles, along with uncertainty intervals. The following code chunk displays the calculation of treated standardized versions.

```
overview_qtt <- qtt(
  overview_causal_fit,
  probs = c(0.50, 0.90, 0.95),
  interval = "credible",
  show_progress = FALSE
)
summary(overview_qtt)
plot(overview_qtt, type = "effect")
```

Conditional causal effets at user-specified covariate profiles are available through the same `predict()` interface, where the `type` argument determines whether quantile, survival, or other summaries are returned. The example below illustrates quantile and survival predictions at a grid of covariate profiles spanning the empirical quartiles of the training predictors.

```
predict(overview_causal_fit,newdata = overview_causal_xgrid,type = "survival",
        y = rep(4, nrow(overview_causal_xgrid)))
```

The bulk-only analogue is `dpmix.causal()`, which keeps the same causal estimand and prediction interface without the GPD tail. For randomized experiments, the same modeling workflow applies without PS augmentation, and the `PS` argument can be set to `NULL`.

4. Data analysis I: Clustering

This Section 4 outlines the clustering process that applies to a spliced conditional mixture model. The goal is to identify latent groups among observational units, as indicated by the mixture model, and summarize clustering uncertainty in a manner that is invariant to labeling. The results will be communicated through (i) the posterior similarity matrix (PSM) and (ii) a representative partition, along with out-of-sample predictions for cluster labels.

4.1. Data and preprocessing

This analysis uses the `Boston`housing data (MASS) package, where each census tract is considered an observational unit. The target variable is median house value (`medv`), and the predictor variables are `lstat`, `rm`, and `nox`. The goal is to identify representative latent groups and then predict which group a new tract belongs to, using out-of-sample predictions from posterior predictive co-clustering evidence. The data is split into a training set and a test set, and the predictor variables are standardized using moments estimated from the training data. These preprocessing tasks are omitted from the code for brevity.

```
library(MASS)

set.seed(123)
```

```
data("Boston", package = "MASS")

dat <- Boston
dat <- subset(dat, select = c(medv, lstat, rm, nox))

n <- nrow(dat)
idx_train <- sample(seq_len(n), size = floor(0.75 * n), replace = FALSE)
train_dat <- dat[idx_train, ]
test_dat  <- dat[-idx_train, ]

# standardize covariates using training set
xvars <- c("lstat", "rm", "nox")
mu_x  <- vapply(train_dat[, xvars], mean, numeric(1))
sd_x  <- vapply(train_dat[, xvars], sd,   numeric(1))

train_dat[, xvars] <- sweep(sweep(train_dat[, xvars], 2, mu_x, "-"), 2, sd_x, "/")
test_dat[, xvars]  <- sweep(sweep(test_dat[, xvars],  2, mu_x, "-"), 2, sd_x, "/")
# report sample sizes
data.frame(
  sample = c("train", "test"),
  n = c(nrow(train_dat), nrow(test_dat))
)


  sample   n
1  train 379
2   test 127
```

Figure 4 provides full-sample descriptive views of Boston house prices (`medv`).

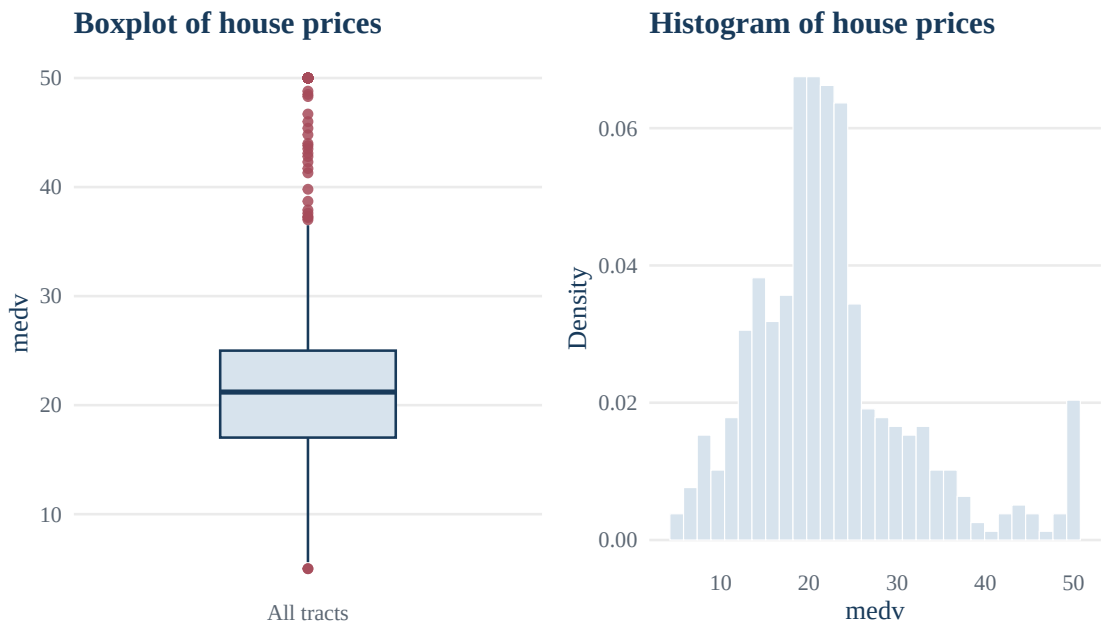

Figure 4: Full-sample distribution of Boston house prices (`medv`): boxplot (left) and histogram (right).

4.2. Model estimation and training cluster summaries

We use a spliced conditional mixture model for the `medv` data with three covariates. Here, we use the option `type="weights"`, where the covariates are used to influence the mixture weights, and the component parameters are shared. This is to demonstrate how the proportion of each cluster changes with the covariates, without changing the clustering framework for the bulk of the response along with the upper tail.

```
fit_clust <- dpmgpd.cluster(
  formula    = medv ~ lstat + rm + nox,
  data       = train_dat,
  kernel     = "lognormal",
  type       = "weights",
  components = 15,
  mcmc       = mcmc_fixed
)
```

Because mixture components are not identifiable up to permutation, the training labels are extracted from the posterior through the PSM and Dahl's least-squares representative partition.

```
z_train <- predict(fit_clust, type = "label")
```

Figure 5 shows the cluster-level response overlay for the five most populated representative clusters, produced directly by the package `plot()` method applied to the training labels object.

```
plot(z_train, type = "summary", top_n = 5)
```

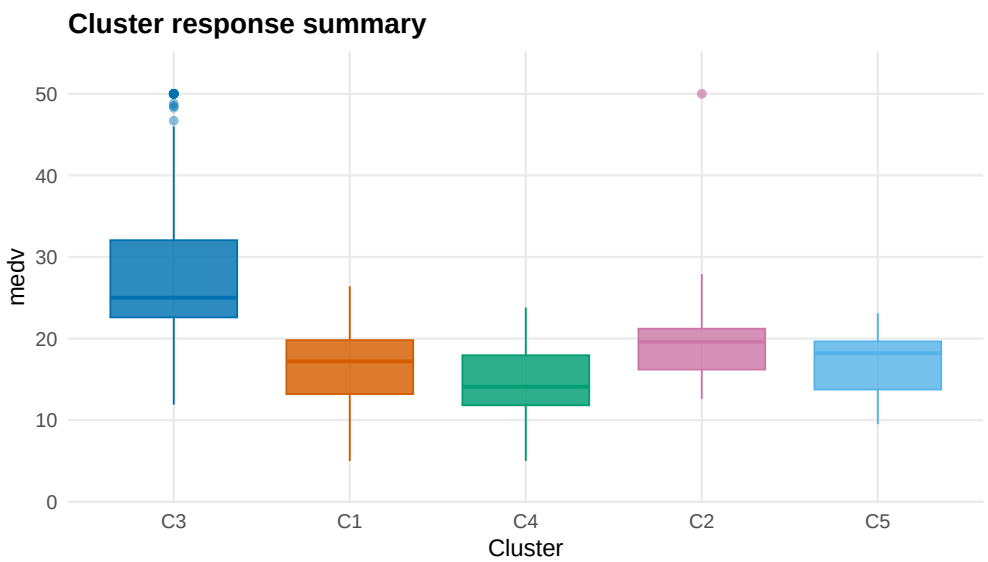

Figure 5: Cluster-level response overlay for the five most populated representative training clusters (`medv` data). Each panel shows within-cluster response summaries produced by the package `plot()` method.

4.3. Prediction for new observations

Cluster assignment for new observations is performed through the same prediction interface. The output is a label in the space of the representative training clusters, obtained by aggregating posterior predictive co-clustering evidence at the cluster level.

```
z_test <- predict(
  fit_clust,
  newdata = test_dat,
  type    = "label"
)
```

Figure 6 shows the distribution of predicted cluster assignments across the test sample, produced by the package `plot()` method applied to the test labels object.

```
plot(z_test, type = "sizes")
```

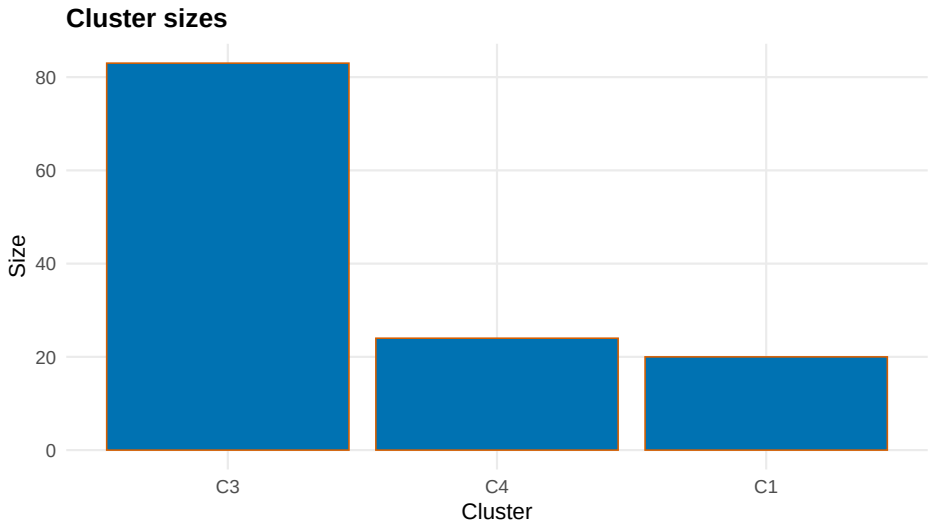

Figure 6: Predicted cluster assignment distribution for the test sample (`medv` data). Each bar corresponds to one representative training cluster; bar heights show the number of test observations assigned to that cluster.

This analysis illustrates the intended clustering workflow: inference is summarized through a representative training partition and its cluster-level descriptive summaries, while new observations are mapped to those same clusters through posterior predictive co-clustering evidence. In practice, the most populated representative clusters are the most interpretable summary object, because they connect posterior partition uncertainty to concrete response and covariate profiles. Section 5 turns from latent partitions to treatment contrasts.

5. Data analysis II: Causal Inference

This application demonstrates causal estimands from a two-arm spliced DPM regression with a GPD tail. In contrast to Section 4, which targets latent partitions, the focus here is on mean and quantile treatment contrasts, including covariate-specific effects and upper-tail behavior.

5.1. Data and estimands

We use the `lalonde` benchmark distributed with MatchIt for the Lalonde job-training application. Here the outcome is 1978 earnings `re78` (USD), the treatment indicator is randomized program assignment `treat`, and the adjustment set includes age, education, race, marital status, degree status, and lagged earnings `re74` and `re75`.

Throughout, we report the four causal estimands in Table 1, using the notation introduced in Section 2.5. The standardized estimands (ATE, QTE) integrate over the empirical distribution of $\{\boldsymbol{x}_i\}_{i=1}^n$, while the conditional estimands (CATE, CQTE) are evaluated at representative Lalonde participant profiles.

The following chunk loads the data and constructs the design matrix used in the causal fit. To stabilize the GPD-linked regression, the model uses centered and scaled versions of the continuous covariates while the outcome remains on the original earnings scale. That preprocessing is kept off the printed page so that the visible code focuses on the package interface.

Figure 7 provides full-sample descriptive views of observed 1978 earnings (`re78`) by treatment assignment.

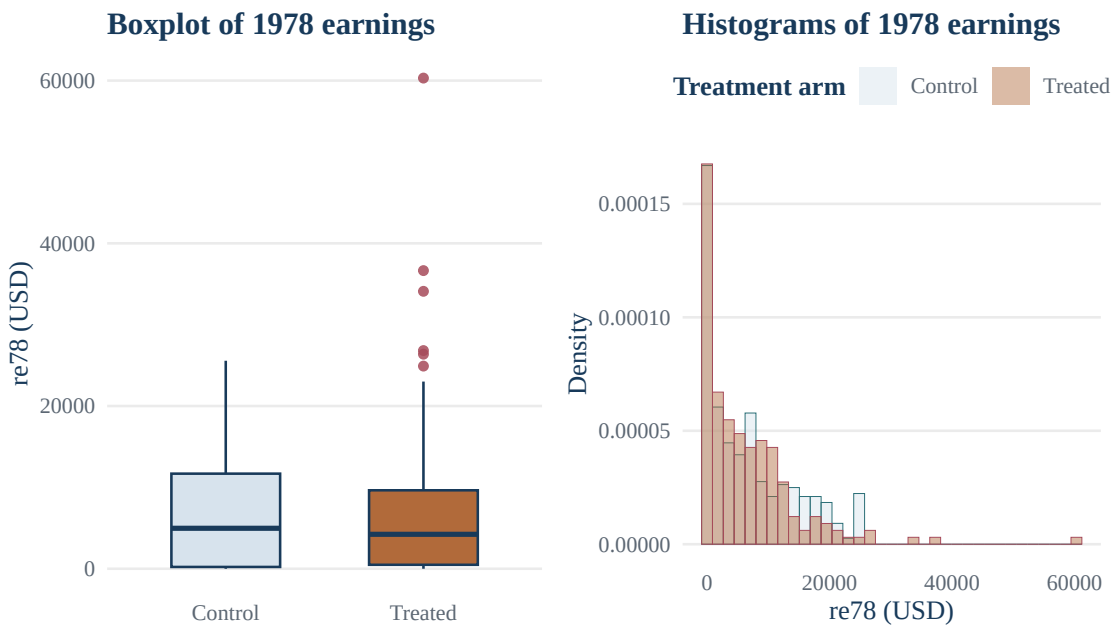

Figure 7: Full-sample distribution of Lalonde 1978 earnings (`re78`): boxplot by treatment arm (left) and overlaid histograms by treatment arm (right).

5.2. Model and fit

Arm-specific conditional outcome distributions $F_a(\cdot \mid \boldsymbol{x})$ are modeled via the spliced bulk–tail construction described in Section 2.3, where

$$f_a(y \mid \boldsymbol{x}) = \sum_j w_{aj}(\boldsymbol{x})\, k(y \mid \Theta_{aj})\, \mathbf{1}(y \le u) + \{1 - F_a(u \mid \boldsymbol{x})\}\, g(y \mid u, \sigma_a, \xi_a)\, \mathbf{1}(y > u).$$

The bulk kernel $k(\cdot)$ is a DPM regression (stick-breaking backend) and the tail density $g(\cdot)$ is GPD with scale $\sigma_a > 0$ and shape ξ_a. The GPD tail index ξ_a controls extremal heaviness (e.g., $\xi_a > 0$ corresponds to heavy tails). This model stabilizes inference for extreme quantiles by coupling nonparametric bulk flexibility with EVT-consistent tail extrapolation.

We fit a two-arm spliced mixture using the Gamma bulk kernel and the same MCMC settings used throughout the manuscript. This is not a definitive substantive analysis; the goal is to show how the package reports standardized and profile-specific Lalonde earnings contrasts under the spliced DPM–GPD workflow.

```
app_lalonde_fit <- dpmgpd.causal(
  x = app_lalonde_y,
  X = app_lalonde_X,
  treat = app_lalonde_A,
  backend = "crp",
  kernel = "gamma",
  components = 10,
  mcmc = c(mcmc_fixed, list(show_progress = FALSE))
)
```

5.3. Posterior summaries

Before reporting treatment contrasts, we summarize posterior arm-specific tail parameters. These tables provide the parameter-level posterior context used in the interpretation of mean-based and quantile-based estimands.

Table 2: Posterior summaries of arm-specific GPD tail parameters for treated and control earnings models.

arm	parameter	mean	sd	q0.025	q0.500	q0.975
treated (A=1)	tail shape	0.248	0.142	-0.118	0.314	0.433
control (A=0)	tail shape	0.069	0.066	-0.024	0.093	0.161

Table 3: Posterior summaries for the arm-specific GPD tail-shape parameter ξ_a.

arm	mean	sd	q0.025	q0.500	q0.975
treated (A=1)	0.248	0.142	-0.118	0.314	0.433
control (A=0)	0.069	0.066	-0.024	0.093	0.161

5.4. Standardized estimands: ATE and QTE

We summarize the overall training effect with ATE and QTE on a main-text quantile grid that reaches into the upper earnings tail without relying on the most unstable extreme quantiles.

```
app_lalonde_ate_fit <- ate(
  app_lalonde_fit,
  interval = "credible",
  show_progress = FALSE
)
```

Table 4: Posterior ATE summary for Lalonde 1978 earnings

estimate	lower	upper
0.01	-0.19	0.299

```
app_lalonde_qte_fit <- qte(
  app_lalonde_fit,
  probs = app_lalonde_taus,
  interval = "credible",
  show_progress = FALSE
)
```

Figure 8 displays two standardized QTE views side by side from the spliced causal model: the effect curve and the treated-versus-control quantile curves with uncertainty intervals.

```
app_lalonde_qte_effect_plot <- plot(app_lalonde_qte_fit, type = "effect")
app_lalonde_qte_arms_plot <- plot(app_lalonde_qte_fit, type = "arms")

grid::grid.newpage()
app_lalonde_qte_layout <- grid::grid.layout(nrow = 1, ncol = 2)
grid::pushViewport(grid::viewport(layout = app_lalonde_qte_layout))
print(
  app_lalonde_qte_effect_plot,
  vp = grid::viewport(layout.pos.row = 1, layout.pos.col = 1)
)
print(
  app_lalonde_qte_arms_plot,
  vp = grid::viewport(layout.pos.row = 1, layout.pos.col = 2)
)
grid::upViewport()
```

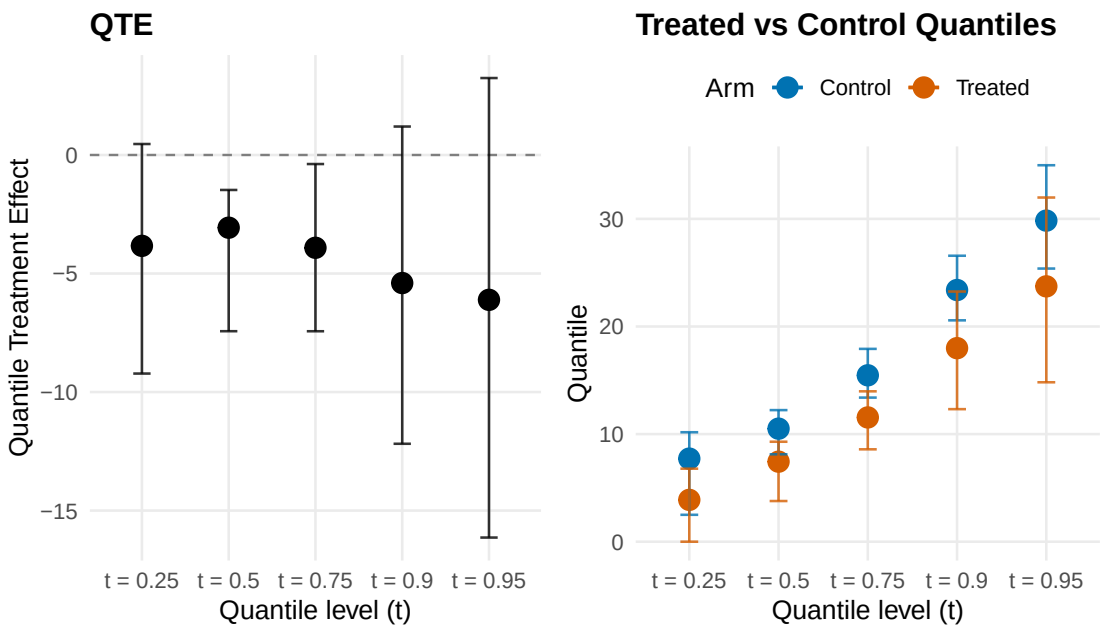

Figure 8: Standardized QTE visual summaries for Lalonde 1978 earnings: effect curve (left) and treated-versus-control quantiles (right), each with pointwise credible intervals.

5.5. Conditional estimands at representative profiles

To illustrate heterogeneity, we evaluate conditional effects at four covariate profiles that differ across all variables. Each profile is constructed on the original data scale using sample quantiles for continuous covariates and observed factor levels for categorical ones. The raw profiles are then standardized using the training-sample centering and scaling constants and mapped through the same `model.matrix()` encoding used in the fit so that factor combinations remain valid. The profile table is shown below; the intermediate design-matrix construction is hidden because it is not part of the package-facing workflow.

Table 5: Representative Lalonde participant profiles used for conditional causal contrasts

profile	age	educ	race	married	nodegree	re74	re75
Profile 1	20	9	black	no	yes	0.00000	0.00000
Profile 2	25	11	hispan	no	no	47.41425	16.47097
Profile 3	32	12	white	yes	no	2705.74760	1389.64860
Profile 4	43	13	black	yes	yes	9383.23560	4201.88700

```
app_lalonde_cate_fit <- cate(
  app_lalonde_fit,
  newdata = app_lalonde_xnew,
  type = "mean",
  interval = "credible",
  show_progress = FALSE
)
```

Figure 9 displays the mean-based CATE estimates across the four representative profiles, produced by the package `plot()` method.

```
plot(app_lalonde_cate_fit)
```

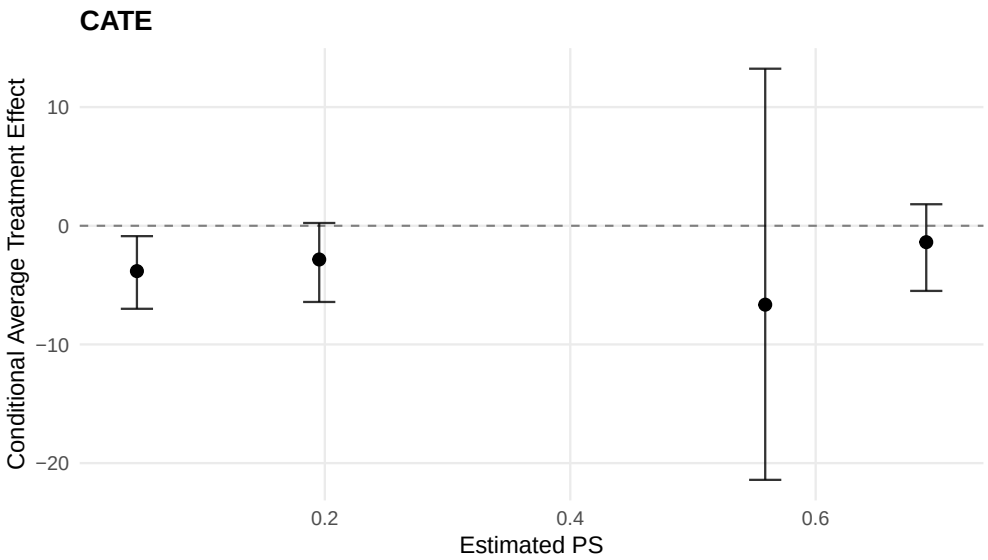

Figure 9: Mean-based CATE estimates for Lalonde earnings at the four representative profiles with pointwise credible intervals.

```
app_lalonde_cqte_fit <- cqte(
  app_lalonde_fit,
  probs = app_lalonde_taus,
  newdata = app_lalonde_xnew,
  interval = "credible",
  show_progress = FALSE
)
```

Table 6 reports CQTE summaries at the four representative profiles on the same quantile grid used in the standardized QTE analysis.

5.6. Results and interpretation

A key advantage of the spliced model is direct posterior inference on the tail index ξ_a by arm. Larger ξ_a indicates a heavier upper tail and greater propensity for extreme earnings realizations in the corresponding arm. Table 2 and Table 3 summarize these posterior parameter estimates.

Because both posterior arm-specific tail-shape means satisfy
hat
$xi_a < 1$, the corresponding arm-specific means exist; therefore the reported ATE and CATE summaries are well-defined in this analysis.

Table 6: Conditional QTE estimates at representative Lalonde participant profiles.

profile	id	index	estimate	lower	upper
Profile 1	1	0.25	-0.902	-5.825	1.032
Profile 1	1	0.50	-3.048	-6.498	-1.121
Profile 1	1	0.75	-0.881	-4.581	1.996
Profile 1	1	0.90	3.084	-2.410	9.621
Profile 1	1	0.95	6.952	0.104	16.955
Profile 2	2	0.25	-0.903	-5.825	0.891
Profile 2	2	0.50	-3.057	-6.405	-1.252
Profile 2	2	0.75	-1.982	-4.181	-0.013
Profile 2	2	0.90	-1.051	-3.696	3.032
Profile 2	2	0.95	-0.031	-3.751	6.596
Profile 3	3	0.25	-0.905	-5.825	0.819
Profile 3	3	0.50	-3.360	-6.718	-1.443
Profile 3	3	0.75	-3.581	-5.435	-1.725
Profile 3	3	0.90	-4.592	-7.754	0.223
Profile 3	3	0.95	-5.092	-10.215	2.652
Profile 4	4	0.25	-0.873	-5.825	2.806
Profile 4	4	0.50	-5.677	-11.538	-0.580
Profile 4	4	0.75	-10.452	-27.958	11.166
Profile 4	4	0.90	-12.909	-51.629	37.293
Profile 4	4	0.95	-12.062	-67.945	67.616

Taken together, the standardized and profile-specific summaries separate two distinct questions: how the program shifts the overall earnings distribution after averaging over the empirical covariate mix, and how that shift changes across substantively different participant profiles. The tail-shape summaries then add an explicitly extremal comparison by showing how much posterior mass each arm assigns to heavier or lighter upper-tail behavior.

6. Discussion

CausalMixGPD combines flexible bulk density estimation, optional EVT-based tail extrapolation, predictor-dependent clustering, and distributional causal inference in one workflow. The same fitted model supports posterior summaries for densities, survival curves, means, restricted means, quantiles, and treatment contrasts, which avoids the fragmented analysis pipelines often required when heavy tails and distributional effects matter simultaneously.

The package is broader than the spliced examples emphasized here. Users can work with bulk-only or spliced models, multiple kernel families on the real line or positive support, SB or CRP backends, and clustering or causal wrappers built on the same predictive machinery. That shared infrastructure makes it easier to move from exploratory subgroup analysis to tail-focused prediction and causal reporting without changing the underlying modeling language.

The first practical limitation is computation. Mixture models with covariate dependence, optional tail components, and arm-specific causal fits are MCMC-intensive, and the cost increases further when users run sensitivity analyses, longer chains, or richer predictor sets.

The second limitation is model tuning. A spliced analysis inherits choices from both the DPM and peaks-over-threshold components, including kernel family, truncation level, threshold specification, and the number of exceedances available for tail learning. When overlap is

weak or exceedances are sparse, posterior uncertainty can be substantial, and for clustering the posterior similarity matrix is often more reliable than any single partition summary.

The third limitation is scope. The current causal implementation is restricted to binary treatments and continuous outcomes under standard consistency, ignorability, and overlap assumptions. Propensity-score augmentation can improve adjustment in observational studies, but it is not a substitute for design, overlap diagnostics, or sensitivity analysis for unmeasured confounding.

References

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